THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE TWO-CHAPTER TEST SET C

Chapters 3 & 4: **The Atomic Foundations of Matter, Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The molecular/formula unit mass of sulfuric acid, H_2SO_4 is (H = 1, C = 12, N = 14, O = 16, Na = 23, Mg = 24, S = 32, Cl = 35.5, Ca = 40): 1

(a) 196 u **(b)** 100 u **(c)** 96 u **(d)** 98 u

2. The molecular/formula unit mass of magnesium chloride, $MgCl_2$ (formula unit mass) is (H = 1, C = 12, N = 14, O = 16, Na = 23, Mg = 24, S = 32, Cl = 35.5, Ca = 40): 1

(a) 95 u **(b)** 93 u **(c)** 97 u **(d)** 190 u

3. Water always contains hydrogen and oxygen in the mass ratio: 1

(a) 8 : 1 **(b)** 1 : 1 **(c)** 2 : 1 **(d)** 1 : 8

4. The chemical formula of aluminium oxide is: 1

(a) AlO **(b)** Al_2O **(c)** Al_2O_3 **(d)** Al_3O_2

5. The chemical formula of calcium carbonate is: 1

(a) $Ca(CO_3)_2$ **(b)** CaC_2O_3 **(c)** Ca_2CO_3 **(d)** $CaCO_3$

6. An atom of aluminium has atomic number 13 and mass number 27. The number of neutrons in it is: 1

(a) 13 **(b)** 14 **(c)** 15 **(d)** 27

7. The electronic configuration (K, L, M, N shells) of the chlorine atom (Z = 17) is: 1

(a) 2, 8, 7 **(b)** 2, 9, 6 **(c)** 2, 8, 8 **(d)** 8, 2, 7

8. The electronic configuration (K, L, M, N shells) of the aluminium atom (Z = 13) is: 1

(a) 2, 8, 3 **(b)** 8, 2, 3 **(c)** 2, 9, 2 **(d)** 2, 8, 4

9. The valency of oxygen (Z = 8) is: 1

(a) 1 **(b)** 3 **(c)** 4 **(d)** 2

10. Acharya Kanada called the smallest indivisible particles of matter: 1

(a) atomos **(b)** quarks **(c)** electrons **(d)** parmanus

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Write the formulae of 2

(a) calcium chloride

(b) aluminium oxide.

12. Calculate the molecular mass of CO_2 and NH_3 (C=12, O=16, N=14, H=1). 2

13. What is valency? How is it related to the valence shell? 2

14. Define isotopes with an example of hydrogen. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 30–40 words each.

15. Calculate the molecular mass of H_2SO_4 and the formula unit mass of $CaCl_2$ (H=1, S=32, O=16, Ca=40, Cl=35.5). 3

16. Chlorine has two isotopes of mass 35 u and 37 u in the ratio 3 : 1. Calculate its average atomic mass. 3

17. Why does NaCl conduct electricity in solution but sugar does not? 3

OR

Why do ionic compounds have high melting points and are usually hard but brittle? 3

18. A bromine sample has isotopes 79 u (49.7 %) and 81 u (50.3 %). Find the average atomic mass. 3

OR

Why is the weighted average atomic mass of chlorine 35.5 u even though no atom has this mass? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) What is nuclear energy? Name one use. 5
(ii) Name the scientist called the Father of the Indian Nuclear Programme.
(iii) How does a nuclear power plant produce electricity?

OR

Case study: A student reacted zinc with dilute hydrochloric acid in an open test tube and found the mass decreased. 5
(i) Name the products.
(ii) Why did the mass decrease?
(iii) How can she modify the set-up to verify the conservation of mass?
(iv) What result should she get?

20. **Case study:** Table of elements: Na (2,8,1), Mg (2,8,2), Cl (2,8,7), Ar (2,8,8). 5
(i) Which has complete valence shell?
(ii) Which will lose electrons?
(iii) Which will gain?
(iv) Predict the formula of the compound of Mg and Cl and explain.

OR

(i) Distinguish between Thomson, Rutherford and Bohr models in a table of three points. 5
(ii) Which discovery showed that atoms are divisible?
(iii) Name the discoverers of electron, proton and neutron.

END OF QUESTION PAPER Total: 40 Marks